

Tariff Impacts on U.S. Microeconomic Outcomes: Evidence from Polysilicon, Steel, and Aluminum Imports

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1 Introduction

Over the past decade, U.S. trade policy has increasingly relied on tariffs as a tool to influence domestic production, supply chain security, and geopolitical leverage. Evidence shows that the tariffs reduced import growth from China and stimulated import growth from other countries; however, the growth rate of overall imports of the tariffed products remained the same (Freund et al., 2024), and import tariffs weakened U.S. export growth (Handley et al., 2025). Beginning with the Section 232 tariffs on steel and aluminum (2018) and continuing through targeted tariffs and restrictions on advanced inputs such as polysilicon, these measures were intended to protect domestic industries, reduce reliance on strategic rivals, and stabilize critical supply chains. However, tariffs also act as taxes on imports and can generate complex microeconomic effects that propagate through prices, quantities, and downstream industries.

This report evaluates the microeconomic impacts of tariffs using three strategically important intermediate goods:

- Poly silicon (a key input for semiconductors and solar photovoltaics)
- Steel products
- Aluminum products

Using monthly U.S. import values from Census HS trade data and Producer Price Index (PPI) series from Federal Reserve Economic Data (FRED) for related downstream industries, this analysis examines how tariffs affect import volumes, domestic prices, and market adjustment dynamics. The figures included in this repository plot monthly import values (bars) against indexed PPI series (lines), allowing for a joint interpretation of quantity and price responses around tariff implementation periods.

The objective is not to estimate causal elasticities, but to provide descriptive, policy-relevant microeconomic inferences that can inform future trade and industrial policy discussions.

2 Data and Methodology

2.1 Data Sources

- U.S. Census Bureau – International Trade Data. Monthly import values by Harmonized System (HS) codes, expressed in nominal USD.
- Federal Reserve Economic Data (FRED) :Producer Price Index (PPI) series corresponding to semiconductor manufacturing, steel mill products, and aluminum-related products. Each PPI series is indexed to January 2018 = 100 for comparability.

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2.2 Selected Products

- **Polysilicon:** Trade data are drawn from Harmonized System six-digit code (HS6) **280461**, classified as *polysilicon (ultra-pure silicon, 99.99%)*. The price proxy used is the Producer Price Index (PPI) for semiconductors, **PCU334413334413**, obtained from FRED.
- **Steel:** Trade data correspond to flat-rolled steel products under Harmonized System four-digit code (HS4) **7208**, classified within iron and steel products. The price proxy is the Producer Price Index for steel mill products, **WPU1017**, from FRED.
- **Aluminum:** Trade data are based on Harmonized System four-digit code (HS4) **7606**, classified as *aluminum plates, sheets, and strip exceeding 0.2 mm in thickness*. The price proxy used is the Producer Price Index for metals and metal products—extruded aluminum shapes (rod, bar, and wire), **WPU10250162**, from FRED.

2.3 Analytical Framework

From a microeconomic perspective, tariffs introduce a wedge between world prices and domestic prices. The expected short-run effects include:

- Higher domestic prices (reflected in rising PPI)
- Reduced import quantities
- Potential substitution toward domestic production or alternative foreign suppliers
- Cost pass-through to downstream industries

As it's too early and appropriate data is still unavailable the cost, past- through to downstream and how domestic production substitute the foreign supplies, I used plotting method to see price and quantity response to tariff. The plots jointly visualize:

- Quantity response: Changes in monthly import values
- Price response: Changes in producer prices faced by domestic firms
- Vertical markers indicate the timing of major tariff announcements or enforcement dates.

3 Poly silicon Tariffs and Semiconductor Supply Chains

3.1 Observed Patterns

U.S. trade policy toward poly silicon tightened substantially in 2025, with tariffs on Chinese imports rising sharply through successive policy actions (Table 1). Increases implemented under Section 301 and subsequent executive action significantly elevated the effective tariff rate, while the initiation of a Section 232 national security investigation introduced uncertainty regarding the future scope and level of protection. Together, these measures reflect an intensified policy focus on strategic supply chains in solar and energy-related manufacturing.

Table 1: U.S. Tariff Measures on Poly silicon

Date Announced	Effective Date	Effective Rate	Notes and Authority
December 2024	January 1, 2025	50%	Section 301 tariff increase on Chinese poly silicon and solar wafers
February 1, 2025	February 4, 2025	+10%	Executive order covering energy-related imports from China
July 1, 2025	Under investigation	TBD	Section 232 national security investigation initiated

The poly silicon plot (Figure 1) shows a pronounced decline in import values following tariff escalation and trade restrictions targeting Chinese-origin inputs. Imports trend downward after the tariff period, with heightened volatility rather than immediate recovery. In contrast, the semiconductor-related PPI initially remains elevated and only gradually declines.

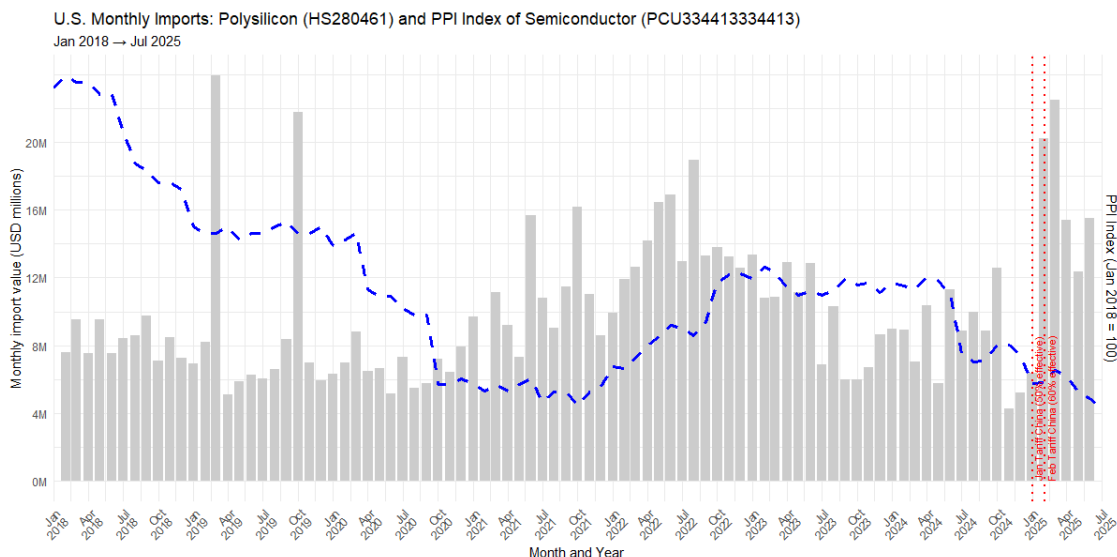


Figure 1: U.S. Monthly Poly silicon Imports and Semiconductor Producer Price Index

3.2 Microeconomic Interpretation

Poly silicon is a highly specialized upstream input with limited short-run substitutes. As a result:

- Inelastic short-run demand: U.S. semiconductor and solar manufacturers cannot easily substitute away from polysilicon, leading to higher effective input costs.
- Price stickiness and delayed adjustment: Even as import volumes fall, PPI remains elevated due to constrained supply and high adjustment costs.
- Cost pass-through: Elevated input prices increase marginal costs for downstream producers, potentially reducing output or increasing final goods prices.

3.3 Welfare and Efficiency Implications

While tariffs may encourage domestic poly silicon investment over the long run, the short- to medium-run effects suggest:

- Higher production costs for strategic industries
- Reduced competitiveness in global semiconductor and solar markets
- Transitional welfare losses borne by downstream producers and consumers

This case illustrates how tariffs on upstream, high-complexity inputs can unintentionally weaken downstream industrial performance.

4 Steel Tariffs and Domestic Price Protection

4.1 Observed Patterns

Steel tariffs imposed under Section 232 have undergone several expansions since their introduction in 2018 (Table 2). After a period of exemptions and country-specific arrangements, the United States reinstated a uniform tariff regime in 2025 and subsequently raised the tariff rate. The later inclusion of additional derivative products broadened the coverage of the policy, extending tariff exposure further along the steel value chain.

Table 2: U.S. Tariff Measures on Steel

Date Announced	Effective Date	Effective Rate	Notes and Authority
March 2018	March 23, 2018	25%	Section 232 tariff imposed on steel imports citing national security concerns
February 2025	March 12, 2025	25%	Termination of country exemptions and implementation of a uniform tariff
June 2025	June 4, 2025	50%	Tariff rate increased for most trading partners
August 2025	August 18, 2025	50%	Expansion of Section 232 coverage to derivative steel products

Following the imposition of Section 232 steel tariffs, steel import values decline sharply, particularly in the immediate post-tariff period (Figure 2). At the same time, the steel PPI rises significantly and remains elevated for an extended period.

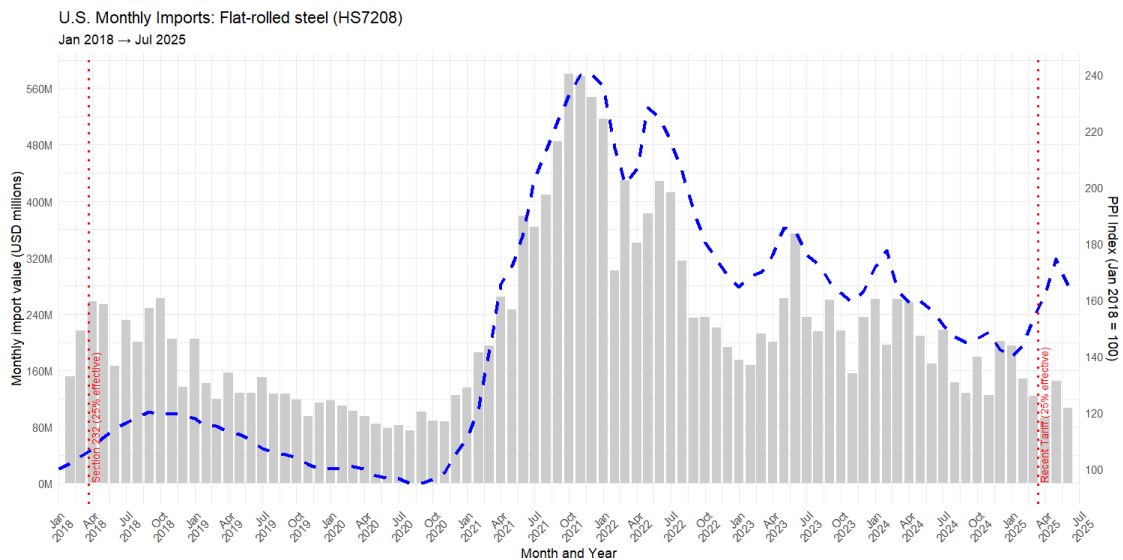


Figure 2: U.S. Monthly Polysilicon Imports and Semiconductor Producer Price Index

4.2 Microeconomic Interpretation

The steel market exhibits characteristics consistent with classic tariff theory:

- Import compression: Higher tariffs reduce foreign supply, shifting demand toward domestic producers.
- Increased domestic prices: Reduced import competition allows domestic firms to raise prices.

- Producer surplus gains: Domestic steel producers benefit from higher prices and increased market share.
- However, steel is a widely used intermediate input, meaning downstream industries face higher costs.

4.3 Downstream Spillovers

Industries such as construction, automobiles, and machinery experience:

- Increased marginal costs
- Reduced output or delayed investment
- Incentives to offshore production of steel-intensive goods

From a microeconomic efficiency standpoint, the gains to protected producers are partially offset by losses to downstream users, consistent with deadweight loss predictions from partial equilibrium models.

5 Aluminum Tariffs and Market Adjustment

5.1 Observed Patterns

Aluminum tariffs followed a trajectory similar to steel under the Section 232 framework (Table 3). Initial measures were reinforced in 2025 through the removal of exemptions and an increase in tariff rates, accompanied by an expansion of coverage to additional downstream products. These changes increased the overall trade restrictiveness affecting aluminum imports while maintaining a largely uniform application across trading partners.

Table 3: U.S. Tariff Measures on Aluminum

Date Announced	Effective Date	Effective Rate	Notes and Authority
March 2018	March 23, 2018	25%	Section 232 tariff imposed on aluminum imports
February 2025	March 12, 2025	25%	Removal of exemptions and reinstatement of uniform tariffs
June 2025	June 4, 2025	50%	Tariff rate increased for most countries
August 2025	August 18, 2025	50%	Expansion of tariff coverage to derivative aluminum products

The aluminum plot shows a more muted decline in import values compared to steel, alongside a moderate but persistent increase in aluminum-related PPI (Figure 3). Over time, import levels partially recover, suggesting supplier substitution rather than sustained import suppression.

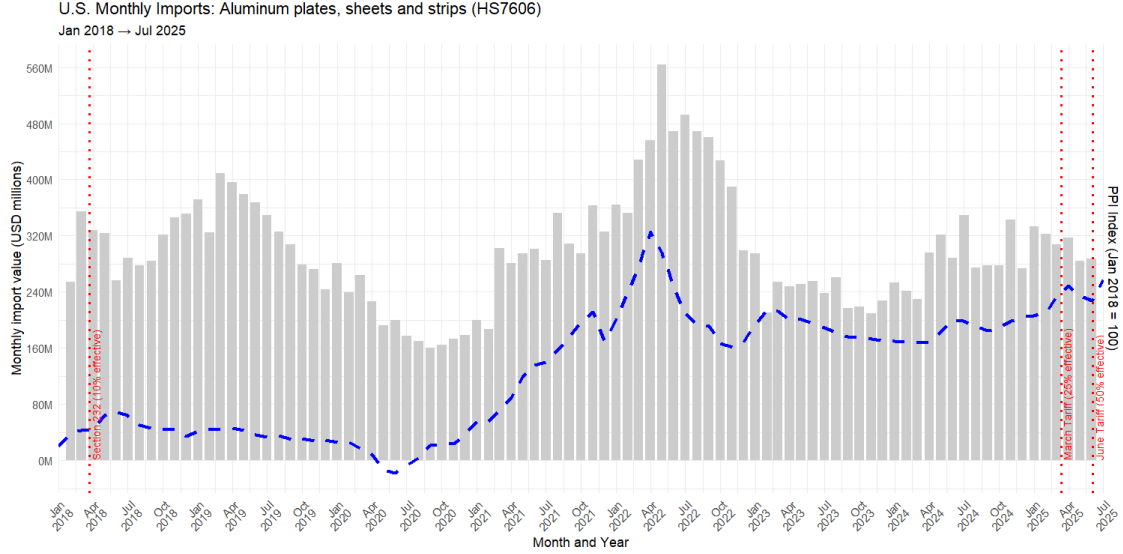


Figure 3: U.S. Monthly Polysilicon Imports and Semiconductor Producer Price Index

5.2 Microeconomic Interpretation

Relative to steel:

- Higher substitutability across suppliers: Aluminum imports can be rerouted from non-tariffed countries more easily.
- Partial pass-through: Prices rise, but less sharply than in steel markets.
- Faster adjustment: Import volumes recover as firms adapt sourcing strategies.

This reflects a more elastic foreign supply curve and greater global market integration.

5.3 Comparative Efficiency

Aluminum tariffs generate smaller but still meaningful welfare losses. The evidence suggests:

- Less effective long-term import suppression
- Persistent but moderate cost increases for downstream users
- Limited ability to permanently insulate domestic producers from global competition

6 Cross-Commodity Comparison

Across the three commodities, tariff impacts differ systematically based on market structure (Table 4).

Table 4: Comparing tariff impacts on Polysilicon, Steel and Aluminum

Commodity	Import Response	Price Response	Key Mechanism
Polysilicon	Sharp decline	Sticky, elevated PPI	Supply chain rigidity
Steel	Large decline	Strong PPI increase	Domestic market power
Aluminum	Moderate decline, recovery	Moderate PPI increase	Supplier substitution

The results align with microeconomic theory: tariffs are most distortionary when demand is inelastic and substitutes are limited.

7 Policy Implications

- Input tariffs have downstream costs: Tariffs on upstream inputs such as polysilicon disproportionately harm strategic downstream industries.
- Producer protection vs. economy-wide efficiency: Steel tariffs demonstrate that producer gains often come at the expense of broader industrial competitiveness.
- Market structure matters: Aluminum markets adjust more efficiently due to greater global substitutability, limiting tariff effectiveness.
- Complementary policies are essential : Tariffs alone are insufficient; investment subsidies, R&D support, and supply chain diversification policies are needed to mitigate short-run welfare losses.

8 Conclusion

This report provides descriptive microeconomic evidence that U.S. tariffs on polysilicon, steel, and aluminum significantly affected import behavior and domestic prices, with heterogeneous outcomes driven by market structure and supply chain complexity. Consistent with the findings of [Freund et al. \(2024\)](#), the results suggest that tariffs altered the composition of U.S. imports—reducing imports from tariff-targeted countries while facilitating substitution toward alternative foreign suppliers—without materially reducing overall import demand for the affected products. At the same time, the observed increases in producer prices and input costs align with evidence in [Handley et al. \(2025\)](#) that import tariffs can weaken export competitiveness by raising production costs for downstream industries.

While tariffs can provide short-run protection to domestic producers, the evidence in this report indicates that such gains are accompanied by higher input costs, distorted resource allocation, and welfare losses borne by downstream users. These microeconomic adjustments help explain why tariffs may fail to deliver sustained improvements in trade balances or industrial competitiveness, even when import sourcing patterns change.

For policymakers, the findings underscore the importance of targeted, data-driven trade policy design. Broad tariffs on intermediate goods risk undermining the very industrial competitiveness they aim to protect, particularly in sectors characterized by complex global supply chains and limited short-run substitutability.

9 Reproducibility and Next Steps

All data sources are publicly available from the U.S. Census Bureau and FRED. Future extensions of this work could include:

- Difference-in-differences estimation with control commodities
- Pass-through elasticity estimation
- Firm-level downstream impact analysis

This repository is intended as a transparent, extensible foundation for further applied trade policy research.

References

- Freund, C., Mattoo, A., Mulabdic, A., and Ruta, M. (2024). Is us trade policy reshaping global supply chains? *Journal of International Economics*, 152:104011.
- Handley, K., Kamal, F., and Monarch, R. (2025). Rising import tariffs, falling exports: When modern supply chains meet old-style protectionism. *American Economic Journal: Applied Economics*, 17(1):208–238.